

Experimental Characterization of Inertial Effects in a Windwall System

Abstract

This study presents an experimental characterization of the inertial response of a windwall system subjected to time-varying operating conditions. Using measured transient responses between discrete velocity states, a transition time matrix is constructed that quantifies acceleration and deceleration times across the full operating range.

The results reveal a strong asymmetry between acceleration and deceleration dynamics, with deceleration times systematically longer due to aerodynamic and flow-mass inertia. Full-range velocity transitions ($2.3 \leftrightarrow 12.4$ m/s) require approximately 0.3 s for acceleration and up to 1.0 s for deceleration, while transitions between rest and maximum velocity extend up to 2.4 s.

These measured time scales impose a strict operational bandwidth on the system. For full-range modulation, the maximum achievable excitation frequency is approximately 0.8 Hz, with a conservative working limit near 0.2 Hz. When oscillating between rest and high-speed conditions, the usable bandwidth reduces to below 0.1 Hz.

These reference values define practical design limits for time-dependent experiments, ensuring physically realizable flow evolution and preventing unrealistic assumptions on temporal controllability.

1 Objective of the Study

The objective of this study is to experimentally characterize the inertial behavior of a windwall system by:

- Quantifying the characteristic times required to transition between different flow velocities.
- Distinguishing between acceleration and deceleration dynamics.
- Identifying the dependence of transition times on the initial flow state.

The final outcome is a transition-time matrix that provides a compact and physically meaningful representation of the system inertia.

2 Experimental Methodology

The system is driven using discrete PWM commands, each associated with a characteristic steady-state flow velocity. For each PWM level, experimental runs consist of three phases:

1. Acceleration from rest to the target velocity,
2. A quasi-steady plateau,
3. Deceleration back towards rest.

The steady-state velocity associated with each PWM command is defined as the *most frequent RPM value* observed during the plateau phase. This definition provides robustness against measurement noise and avoids bias introduced by transient overshoots.

Transition times are computed using interpolated crossing points between discrete RPM samples, ensuring that non-continuous measurements do not artificially quantize the dynamics.

For each pair of velocities, transition times are averaged when multiple realizations are available.

3 Transition Time Matrix

The experimental results are summarized in a transition time matrix, whose entries represent the time required to transition between two flow velocities.

- The upper triangular part corresponds to deceleration processes.
- The lower triangular part corresponds to acceleration processes.
- Axes are expressed in terms of physical flow velocity associated to PWM commands.

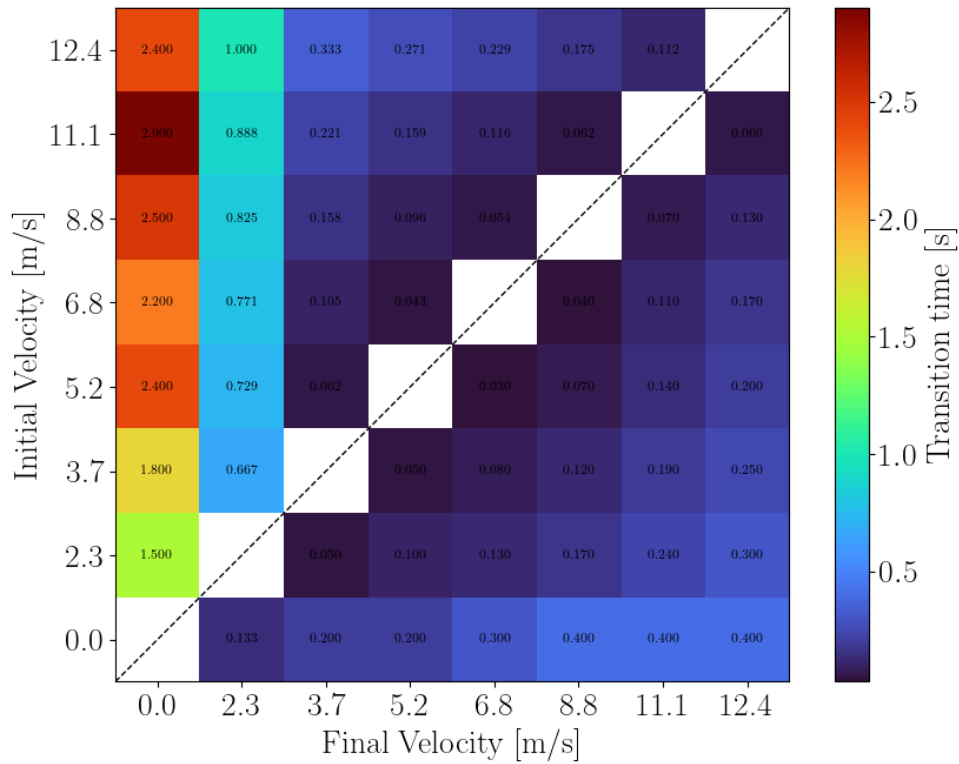


Figure 1: Transition time matrix showing deceleration (upper triangle) and acceleration (lower triangle) times between different flow velocities.

4 Key Findings

The experimental results lead to the following conclusions:

Asymmetric Dynamics

Acceleration and deceleration dynamics are strongly asymmetric. For single-step transitions between adjacent operating points, the median acceleration time is approximately 0.05 s, whereas the median deceleration time increases to 0.063 s. This asymmetry becomes more pronounced for larger velocity changes: over the full operating range ($2.3 \leftrightarrow 12.4$ m/s), acceleration requires approximately 0.30 s, while deceleration requires up to 1.0 s. This systematic difference indicates that the system inertia is dominated by aerodynamic and flow-mass effects rather than by actuator response.

State-Dependent Behavior

The time required to reach a given target velocity depends on the initial flow state. For instance, transitions from rest to the maximum operating velocity require approximately 0.40 s, whereas the return from maximum velocity to rest requires up to 2.4 s. This demonstrates that the transient response is state-dependent and cannot be characterized by a single time constant.

Dynamic Memory and Hysteresis

For identical velocity pairs, acceleration and deceleration times are not equal. This directional dependence reveals the presence of dynamic hysteresis, indicating that the transient response depends not only on the target velocity but also on the direction and history of the transition. Consequently, the dynamics cannot be accurately described by a memoryless or symmetric model.

Operational Bandwidth Limitation

The transition time matrix directly defines the maximum achievable rate of change in flow velocity for periodic or time-dependent modulation. For a full-range oscillation between 2.3 and 12.4 m/s, the combined acceleration–deceleration cycle time is approximately 1.3 s, yielding a theoretical maximum excitation frequency of $f_{\max} \approx 0.77$ Hz. A conservative and practically realizable operating frequency is therefore $f_{\text{work}} \approx 0.23$ Hz. When oscillating between rest and the maximum velocity, the effective bandwidth is further reduced to $f_{\text{work}} \approx 0.11$ Hz. Operating above these frequencies does not lead to faster flow modulation but instead results in incomplete transitions, reduced velocity amplitudes, and distorted temporal waveforms, as the flow cannot physically follow the imposed command and becomes dynamically decoupled from the input signal.

Table 1: Reference transition times and operational frequency limits of the windwall system.

Transition type	Characteristic time [s]	$f_{\max}$ [Hz]	f_{work} [Hz]
Adjacent velocity step (acceleration)	~ 0.05	~ 10	~ 3
Adjacent velocity step (deceleration)	~ 0.06	~ 8	~ 2.5
Full range ($2.3 \leftrightarrow 12.4$ m/s)	~ 1.3 (cycle)	0.77	0.23
Rest $\leftrightarrow$ max velocity	~ 2.8 (cycle)	0.36	0.11

Here, $f_{\max}$ denotes the theoretical maximum excitation frequency based on measured transition times, while f_{work} represents a conservative operational limit including a safety margin.

5 Conclusions

This study demonstrates that the windwall system exhibits significant inertial effects that govern its transient response. Typical response times for small velocity adjustments are on the order

of 0.05–0.06 s, while large-range transitions require up to 1.0 s for deceleration and 2.4 s when returning to rest.

The experimentally derived transition time matrix provides a compact and practical tool for:

- Designing physically realizable control strategies constrained by system inertia,
- Defining admissible excitation frequencies for time-dependent forcing,
- Developing reduced-order or predictive dynamic models that incorporate hysteresis and state dependence.

In particular, the matrix enables a direct estimation of the operational bandwidth of the system, which is limited to approximately 0.2 Hz for full-range velocity modulation and below 0.1 Hz for rest-to-maximum oscillations. Neglecting these inertial constraints may lead to unrealistic assumptions about flow controllability, temporal resolution, and achievable dynamic forcing in experimental applications.